Yue Wu

Division of Applied Mathematics, Brown University, Box F, Providence, RI 02912, USA
Email: yue_wu3@brown.edu URL: https://yuewu2002.github.io

Education

Brown University, Division of Applied Mathematics — Providence, RI, USA
Ph.D. Candidate in Applied Mathematics. Advisor: *Prof. Chi-Wang Shu* — 08/2023–05/2027 (expected)
M.Sc. in Applied Mathematics. GPA: 4.0 / 4.0 — 08/2023–05/2025

University of Science and Technology of China, School of the Gifted Young — Hefei, Anhui, China
B.Sc. in Information & Computational Science. Rank in major: 1st / 43 — 09/2019–06/2023

Research Interests

- **Numerical Methods for PDEs:** discontinuous Galerkin (DG), finite element, spectral element, and finite difference and finite volume weighted essentially non-oscillatory (WENO) methods
- **Scientific Computing:** low-rank method, PDE-constrained optimization, parallel solver development and machine learning

Publications and Preprints

1. **Y. Wu** and C.-W. Shu. Unconditionally L^2-stable explicit–implicit–null filtered Runge–Kutta methods for general nonlinear degenerate parabolic equations. Submitted to *Numer. Math.*
2. **Y. Wu** and C.-W. Shu. Energy stability of explicit–implicit–null time discretizations for nonlinear degenerate parabolic equations with state-dependent coefficients. Submitted to *SIAM J. Numer. Anal.* arXiv:2610.34539
3. **Y. Wu**, S. Becker, J.-M. Qiu, and X. Zhang. Nonnegative low-rank matrix correction under an orthogonality constraint in conservative Vlasov simulations. Submitted to *J. Sci. Comput.* arXiv:2607.26272
4. **Y. Wu** and C.-W. Shu. A positivity preserving and entropy stable nodal discontinuous Galerkin scheme for ideal MHD. Submitted to *Commun. Math. Stat.* arXiv:2604.23885
5. **Y. Wu** and C.-W. Shu. Finite difference alternative WENO schemes with Riemann invariant-based local characteristic decompositions for compressible Euler equations. *J. Comput. Phys.*, 537:114104, Sept 2025. DOI: 10.1016/j.jcp.2025.114104. MR4912873.
6. **Y. Wu** and Y. Xu. A high-order local discontinuous Galerkin method for the p-Laplace equation. *Beijing J. of Pure and Appl. Math.*, 2(1):373–422, Apr 2025. DOI: 10.4310/BPAM.250415002006.

Research Experience

1. **Explicit–Implicit–Null (EIN) Time Stepping for Nonlinear Degenerate Parabolic Equations**
   Brown University. Supervisor: *Prof. Chi-Wang Shu* — 10/2026
   - Proved unconditional energy stability of EIN methods for nonlinear diffusion with gradient flow structure.
   - Developed novel unconditionally L^2-stable EIN filtered Runge–Kutta for general nonlinear diffusion.
2. **Structure-Preserving Discontinuous Galerkin Schemes for Magnetohydrodynamics (MHD)**
   Brown University. Supervisor: *Prof. Chi-Wang Shu* — 03/2025–04/2026
   - Investigated all types of time-explicit structure-preserving conservative schemes for ideal MHD.
   - Developed a nodal discontinuous Galerkin scheme that combines positivity preservation and entropy stability for the first time in literature.
3. **Efficient A-WENO Methods for Compressible Euler Equations**
   Brown University. Supervisor: *Prof. Chi-Wang Shu* — 09/2024–02/2025
   - Investigated the effect of transform variables in local characteristic decompositions on A-WENO performance and proved the optimal practice.
   - Implemented A-WENO methods using a carefully selected set of Riemann invariants to reduce computational cost.
4. **Discontinuous Galerkin Methods for the p-Laplace Equation**
   Bachelor's thesis at USTC. Supervisor: *Prof. Yan Xu* — 12/2022–06/2023
   - Proved rigorous a priori error estimates for a high order local DG scheme.
   - Developed an efficient preconditioned gradient descent method.
5. **Positivity-Preserving Conservative Low Rank Methods for Vlasov Dynamics**

Purdue University (remote). Supervisor: *Prof. Xiangxiong Zhang* 06/2022–08/2022

- Developed multiple nonnegative low rank correction algorithm under orthogonality constraints using convex optimization and tangent space accelerated projection.
- Incorporate the algorithm into the dynamic low rank Vlasov solver as a positivity preserving conservative limiter to obtain robust and physically meaningful simulation.

6. **Numerical Simulation of Plasma Equilibrium Evolution in Nuclear Fusion**
   USTC undergraduate research project. Supervisor: *Prof. Mengping Zhang* 06/2021–05/2022
   - Developed a parallel hybrid finite difference-pseudo spectral code for resistive MHD in toroidal geometry, and performed long-time simulation of resistive tearing mode instability in tokamaks.
   - Validated the results with researchers from the Institute of Plasma Physics, CAS, and against those from existing open-source codes.

Teaching Experience

1. TA @ Brown: *APMA 0355—Applied ODEs with Theory* Fall 2026
2. TA @ Brown: *APMA 1655—Intro. to Prob. & Stats. with Theory* Spring 2026
3. TA @ Brown: *APMA 0160—Intro. to Sci. Computing* Fall 2025
4. TA @ Brown: *APMA 1650—Stat. Inference I* Spring 2025
5. TA @ Brown: *APMA 1210—OR: Deterministic Models* Fall 2024
6. TA @ USTC: *Computational Methods B* Spring 2022

Contributions and Services

- Open-Source Contributor, LDF-RKDG implementation for GP-MHD systems MFEM v4.11
- Journal Reviewer, *J. Comput. Phys.*, *J. Sci. Comput.*, and *Comput. Math. Appl.* since 2025
- Poster Session, the 2024 International Congress of Basic Science (ICBS), Beijing, China 07/2024

Professional Skills

- **Programming:** C++, MATLAB, Python, Fortran 2008, MPI, OpenMP
- **Softwares:** NGSolve, FEniCS, MFEM, Mathematica, Keras, Tecplot
- **Dev Tools:** Linux, Git, Jupyter, Codex, LaTeX, Markdown
- **Language:** Mandarin Chinese, English

Honors and Awards

- Howard and Jan Swearer Graduate Fellowship AY 2023–2024
- USTC Outstanding Undergraduate Award 06/2023
- "Chia-Chiao Lin" Gold Medal (Nation Top 1) in Applied and Computational track & Team Silver Medal & Excellence Prize in Analysis and PDEs track, the 14th S.-T. Yau College Student Mathematics Contest 06/2023
- Gold Prize, USTC Outstanding Student Scholarship 10/2022
- Excellence Prize in Analysis and PDEs track, the 13th S.-T. Yau College Student Mathematics Contest 08/2022
- China National Scholarship 12/2021
- Second Prize, the 13th Chinese Mathematics Competitions 12/2021
- China National Scholarship 12/2020
- Third Prize, USTC Freshman Scholarship 09/2019

Extracurricular Activities

- USTC road cycling team member 09/2019–06/2023

last update: November 1, 2026